

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	49.0 / Female
Department	General Surgery
Diagnosis	Bladder infection
UTI Classification	Complicated UTI
Sample Type	Suprapubic Aspirate

Clinical Presentation

Chief Complaints: Lower abdominal pain;Dysuria

Comorbidities: Diabetes

Surgical History: Hysterectomy

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	32.0	%	20 - 40
Wbc	9900.0	cells/ μ L	4000 - 11000
Polymorphs	66.0	%	40 - 75
Crp	14.0	mg/L	< 10
Rft Serum Creatinine	1.2	mg/dL	0.6 - 1.3
Serum Uric Acid	4.6	mg/dL	3.5 - 7.2
Blood Urea	31.0	mg/dL	7 - 20
Cue Pus Cells	16.0	/HPF	0 - 5
Epithelial Cells	4.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	2.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Candida albicans
Bacteria Type	Fungus (Yeast; the most common cause of candiduria)

Antibiotic Susceptibility Profile

Predicted Sensitive	Amphotericin B, Voriconazole
Predicted Resistant	Fluconazole

Recommended Therapy

Amphotericin B (liposomal formulation)

Dosage: 3 mg/kg IV once daily (e.g., 210 mg for a 70 kg adult) infused over 2-4 hours. Adjust dose if serum creatinine rises >1.5 mg/dL.

Precautions: Renal toxicity: monitor serum creatinine and electrolytes (K⁺, Mg²⁺) daily; ensure adequate hydration. Diabetes: monitor blood glucose as amphotericin can cause hyperglycemia. Use liposomal form to reduce nephrotoxicity. No dose adjustment needed for mild renal impairment (Cr 1.2 mg/dL).

Rationale: Candida albicans is sensitive to Amphotericin B and the organism is resistant to Fluconazole. Amphotericin B provides rapid fungicidal activity and is appropriate for a complicated urinary tract infection in a diabetic patient.

Voriconazole

Dosage: Loading dose 6 mg/kg IV q12h for 2 doses, then maintenance 4 mg/kg IV q12h (≈200 mg PO q12h if switched to oral). Adjust to 3 mg/kg IV q12h if hepatic impairment present.

Precautions: Hepatic function: check LFTs before and during therapy; dose reduce if AST/ALT >5× ULN. Visual disturbances and photosensitivity are common; advise patient to avoid bright light. Monitor for drug-drug interactions (e.g., CYP450 substrates). No dose adjustment required for current renal function.

Rationale: Voriconazole is another agent to which Candida albicans is predicted to be sensitive. It offers an oral option for step-down therapy after initial IV treatment and is effective for complicated candiduria, especially when fluconazole resistance is present.

Antibiotic Drug History

Clinical Summary

Infection: Complicated urinary tract infection (interstitial cystitis) of the bladder in a 49-year-old diabetic female.

Predicted pathogen: *Candida albicans* (fungal candiduria).

Resistance/Sensitivity profile

- Resistant to fluconazole (previous exposure).
- Sensitive to amphotericin B (liposomal) and voriconazole.

Recommended therapy

1. **Amphotericin B (liposomal), 3 mg/kg IV once daily** (≈210 mg for a 70-kg adult) infused over 2-4 h.

Rationale: Provides rapid fungicidal activity and is the drug of choice for fluconazole-resistant candiduria in a complicated UTI, especially in a diabetic patient.

2. **Voriconazole** as step-down or adjunct: loading 6 mg/kg IV q12 h × 2 doses, then 4 mg/kg IV q12 h (≈200 mg PO q12 h if switched).

Rationale: Oral option for continuation after initial IV control; effective against fluconazole-resistant *C. albicans*.

Key precautions / monitoring

- **Renal:** Liposomal amphotericin B is less nephrotoxic, but monitor serum creatinine and electrolytes (K⁺, Mg²⁺) daily; hydrate adequately. No dose reduction needed for current Cr 1.2 mg/dL.
- **Metabolic:** Amphotericin B may worsen hyperglycemia; check blood glucose frequently.
- **Hepatic:** Voriconazole requires baseline and periodic LFTs; reduce dose if AST/ALT >5× ULN.
- **Other:** Voriconazole can cause visual disturbances, photosensitivity, and interacts with many CYP450 substrates-review concomitant meds.

Overall plan: Initiate liposomal amphotericin B promptly; transition to oral voriconazole when clinically stable, while closely monitoring renal function, electrolytes, glucose, and liver enzymes. This regimen addresses the fluconazole resistance and provides effective coverage for the complicated candiduria.